Will YOU HELP US

mobilise the abundance of

Functional Genomics Data

for FAIR reuse in the

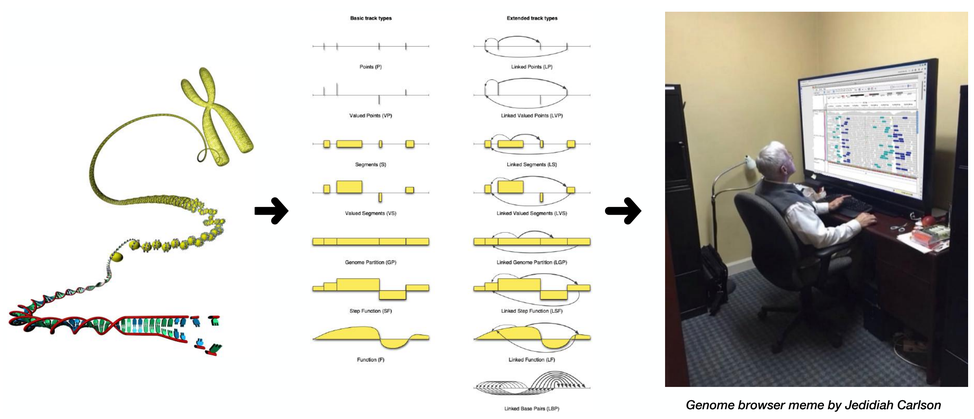
AGE of Pangenomes and AI?

Sveinung Gundersen [1], Joshua Baskaran [1], Federico Bianchini [1], Sanjay Boddu [2], Jeanne Cheneby [1], Finn Drabløs [3*], Nazeefa Fatima [1], José M. Fernández [4], Ahmed Ghanem [1*], Matúš Kalaš [5], Radmila Kompova [1*], Kieron Taylor [2*], Dmytro Titov [1*], Daniel Zerbino [2*], Salvador Capella-Gutiérrez [4], Peter Harrison [2], Eivind Hovig [1,6]

[1] Centre for Bioinformatics / ELIXIR Norway, University of Oslo (UiO), Oslo, Norway
[2] EMBL's European Bioinformatics Institute (EMBL-EBI), Hinxton, United Kingdom
[3] Dept of Clinical and Molecular Medicine, Norwegian University of Science and Technology (NTNU), Trondheim, Norway
[4] Spanish National Bioinformatics Institute (INB) / ELIXIR Spain, Barcelona Supercomputing Centre (BSC), Barcelona, Spain
[5] Department of Informatics / ELIXIR Norway, University of Bergen (UiB), Bergen, Norway
[6] Department of Tumor Biology, Institute for Cancer Research, Oslo University Hospital (OUH), Oslo, Norway
[*] Past project member

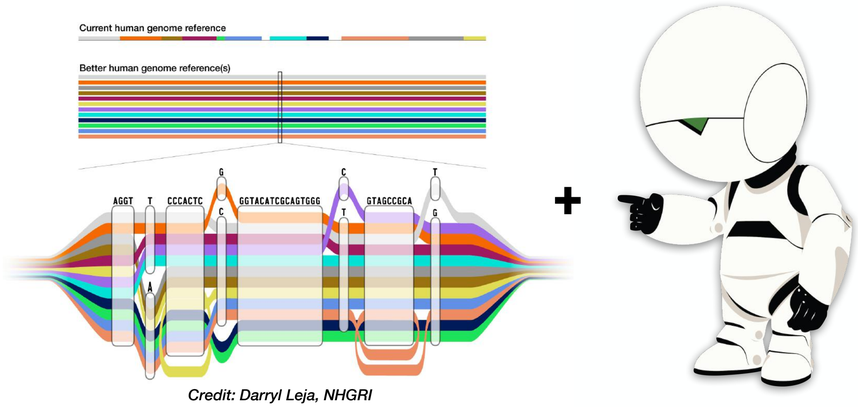
FANTOM 1995 –> | ENCODE 2003 –> | International Cancer Genome Consortium 2008 –> | BLUEPRINT epigenome 2011 – 2017 | ROADMAP epigenomics PROJECT 2008 – 2018 | PCAWG PanCancer Analysis of WHOLE GENOMES 2014 –> | The Cancer Genome Atlas 2005 – 2014 | FAANG Functional Annotation of Animal Genomes 2015 –>

Immense resources have gone into the generation of data that relates genomic positions to functional and structural aspects, in large consortia such as ENCODE, ICGC, FANTOM and FAANG, as well as in smaller research projects.



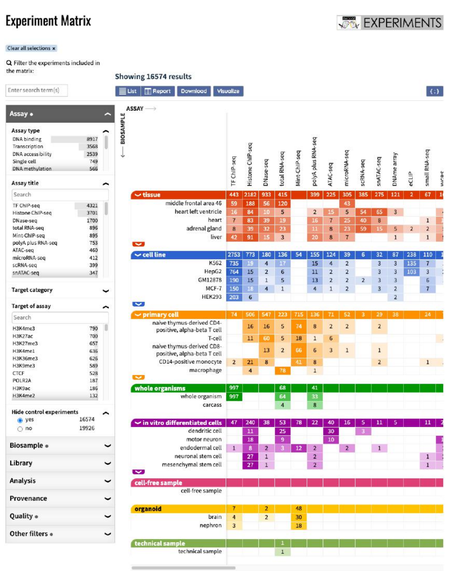
Genome browser meme by Jedidiah Carlson

Functional genomics data are typically provided as tracks for visualisation in genome browsers, or made downloadable for non-visual analysis. The use of reference genomes as one-dimensional coordinate systems constitutes a powerful unified model for data analysis. Interestingly, the condensed nature of coordinate-based data is ideally suited for data–driven approaches!



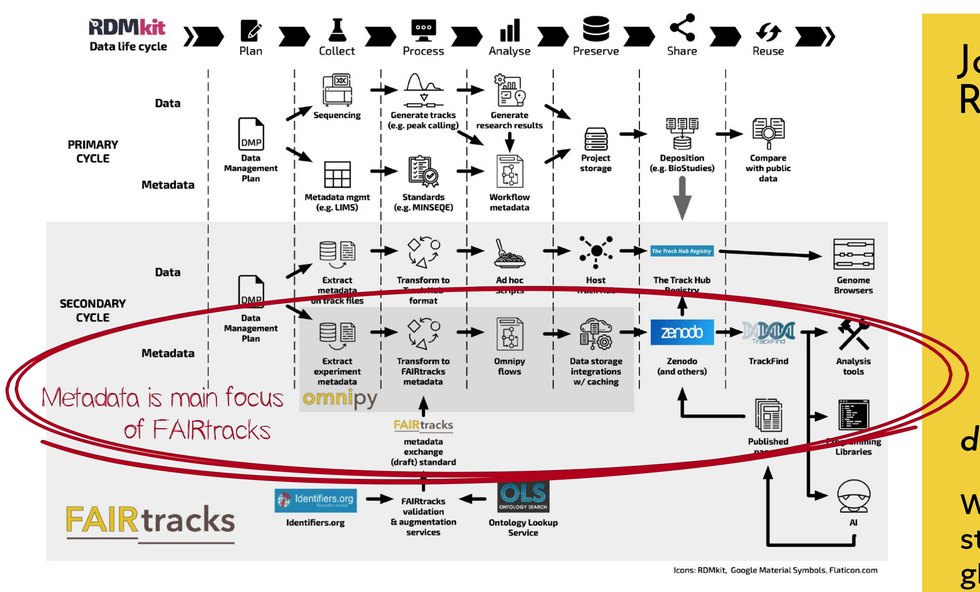
Credit: Darryl Leja, NHGRI

Pangenomes challenge the concept of one-dimensional reference genomes. While a consensus has not yet formed on how to annotate pangenomes, the value of reusing the abundance of existing functional genomics data in this new paradigm is clearly evident. Moreover, data–driven discovery by applying AI to functional annotations still holds considerable untapped potential. To these ends, FAIR (Findable, Accessible, Interoperable, Reusable) metadata remains key.



Functional genomics data are often provided through dedicated data portals, such as the ENCODE data portal (above). However, the metadata are provided according to distinct models and interfaces. Additionally, data from smaller research projects are often available only with limited metadata.

In practice, it is often painstakingly difficult for researchers to locate relevant data for their specific analytical context!



Through European infrastructure funds (ELIXIR), we developed the prototypic FAIRtracks metadata exchange standard. We also developed a fully functioning end-to-end ecosystem of services, including metadata conversion, validation, and categorical search.

UNIVERSITY OF OSLO | BSC Barcelona Supercomputing Center | EMBL-EBI | INB | elixir NORWAY | elixir SPAIN | elixir Recommended Interoperability Resources | RDA TIGER | eosc

Project / community website: FAIRtracks.net

Join our initiative in the Research Data Alliance (RDA):

New Global Support Services for RDA Working Groups

"FAIRification of Genomic Tracks Working Group"

Enabling data-driven life science through uniform and granular discovery of sequence annotation data!

We aim to evolve our solutions into a global standard through the RDA and invite the global community to help us to:

1. Allow for new data types and content
2. Improve interoperability with other solutions
3. Increase adoption by new data providers and downstream analysis software
4. Cater for new user communities

Are you a:
- data producer
- tool developer
- biocurator
- domain expert, or
- analytical end user

...and interested in taking part in the FAIR/Open Science community?

Fill out our community survey!
https://nettskjema.no/a/fairtracks-wg

Read more about the initiative
https://tinyurl.com/bddsua24